Shayan Poigai

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Education

Santa Clara University — B.S. in Computer Science Expected Fall 2027
Major GPA: 4.0

Work Experience

Co-Founder & Software Engineer May 2026 – Present
Adorus (adorusjewels.com)

- Sole developer on a 4-person founding team, translating co-founder and stakeholder feedback into a full-stack jewelry e-commerce platform built end-to-end, including a RESTful API backing the storefront, across 6 planned development phases (pre-launch).
- Designed and shipped an AI virtual try-on feature rendering a photorealistic preview of a customer wearing the exact product via Google Gemini, served through authenticated REST endpoints; benchmarked 5+ image-generation approaches on identity preservation, fidelity, latency, and cost (~$0.05/image).
- Implemented a custom AWS build over Shopify, self-hosting a custom RESTful platform for ~$40/month vs. ~$250+/month for an equivalent plan plus paid apps — an ~80% (~$2,500/year) cut in recurring platform costs.
- Built a responsive React (Vite) frontend on Java Spring Boot RESTful services and PostgreSQL (AWS RDS) across 5 AWS services (EC2, RDS, S3, CloudFront, Cognito), with GitHub Actions CI/CD, 33 Flyway migrations, and 35+ RESTful API endpoints; integrated Stripe for PCI-compliant checkout and Cognito auth.
- Diagnosed a tax-accounting bug by sourcing authoritative totals from Stripe's REST API, backed with 3 regression tests; authored a blog post on human oversight of AI coding tools after a production migration incident.

Software Engineering Intern June 2026 – Present
HerbsPro

- Implementing backend features across the e-commerce stack.

AI/ML Research Assistant Dec 2025 – June 2026
Santa Clara University

- Collaborated with a research team to co-author a peer-reviewed paper as primary writer, accepted at HIBIBI 2026 (ASONAM-collocated), publishing with Springer.
- Built an NLP/data pipeline analyzing 5,000+ scraped posts from Twitter, Reddit, and YouTube to study public sentiment toward LLMs as a therapeutic tool, applying BERTopic, LDA, and NMF topic modeling.
- Engineered Pandas preprocessing to normalize unstructured text, reducing noise in high-dimensional social media datasets.
- Produced Matplotlib / Seaborn analytics on shifting attitudes toward AI mental-health support.

CS Teaching Assistant & Algorithms Grader Sept 2025 – June 2026
Santa Clara University

- Mentored 40+ students in C++ (memory management, pointers, OOP) and debugged logic errors live using GDB.
- Graded Theory of Algorithms coursework for 40+ students across 2 sections, evaluating runtime analysis, greedy, divide-and-conquer, and dynamic-programming solutions.

Projects

Interactive 3D Portfolio | *Next.js, React, Three.js, TypeScript* Live 2026

- Designed and built an interactive, real-time 3D web experience (solar-system navigation, animated shaders, WebGL scenes) using Three.js and React, with a focus on responsive UI and smooth 60fps rendering.
- Rapidly prototyped and iterated on multiple UI concepts; deployed server-rendered on a custom domain with full SEO optimization.

Disease Tracker (AWS Hackathon) | *React, FastAPI, AWS Bedrock, DynamoDB* Github Oct 2025

- Built with a team of 4 at an AWS hackathon; developed a FastAPI REST API serving real-time disease metrics and risk assessments with low-latency responses.
- Engineered an AWS Bedrock AI risk-scoring pipeline and designed DynamoDB schemas for scalable data ingestion.

Kuhn Poker vs Quantum | *Qiskit, IBM Quantum, FastAPI, React, Vercel, Render* Github, Live June 2026

- Built a full-stack game where a 12-parameter, 6-qubit variational quantum circuit (superposition + entanglement) derives the opponent's mixed strategy, deployed on Vercel and Render.
- Ran the circuit on real IBM quantum hardware (e.g., ibm_fez, ibm_mukesh); on a 4,096-shot run, real-chip probabilities matched simulation within ~1.7 percentage points on average (max ~4), the gap attributable to quantum noise.

Agentic Restaurant RAG | *Python, ChromaDB, Ollama, MCP, RAG* Github July 2025

- Built a RAG system answering natural-language restaurant questions via ChromaDB semantic search.
- Exposed data as agent tools over MCP; tested on OpenWebUI with Ollama.

Technical Skills

Languages: C++, JavaScript, Python, Java
Frontend & UI: React.js, Next.js, Three.js (WebGL), Node.js, Vite, responsive UI, REST/RESTful APIs
Systems & Tools: Linux/Unix, C++, GDB, Docker, Git, CI/CD (GitHub Actions), Spring Boot, FastAPI
Cloud & Data: AWS (EC2, S3, CloudFront, Cognito, Bedrock), PostgreSQL, DynamoDB
AI / ML: RAG, MCP, NLP, Ollama, NumPy, Pandas, topic modeling (BERTopic, LDA, NMF)
CS Fundamentals: Data Structures and Algorithms, Object-Oriented Programming